

Milestone 3: Machine Learning Model Development and Optimization

Sales and Quantity Forecasting System

Executive Summary

This report details the development and optimization of machine learning models for predicting weekly sales and quantity across three product categories: Technology, Furniture, and Office Supplies. Our approach employed category-specific modeling to account for distinct business patterns, achieving significant improvements in forecasting accuracy.

Key Achievements:

- 15-25% improvement in accuracy using category-specific models vs unified approach
 - Gradient Boosting emerged as the best-performing algorithm across all categories
 - Successful implementation of dual prediction strategy (Sales + Quantity)
 - Comprehensive hyperparameter tuning using GridSearchCV
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Multi-Model Architecture

Why Separate Category Models?

- **Varying Price Distributions:** Technology (\$100-\$2000) vs Office Supplies (\$1-\$100)
- **Different Seasonal Patterns:** Technology has strong holiday peaks
- **Unique Sales Velocities:** Furniture has lower frequency, higher value
- **Distinct Customer Behaviors:** Each category attracts different buying patterns

Feature Engineering

Temporal Features

- Day of week, month, quarter indicators
- Holiday and special event flags
- Rolling averages (7-day, 30-day)

- Year-over-year growth metrics

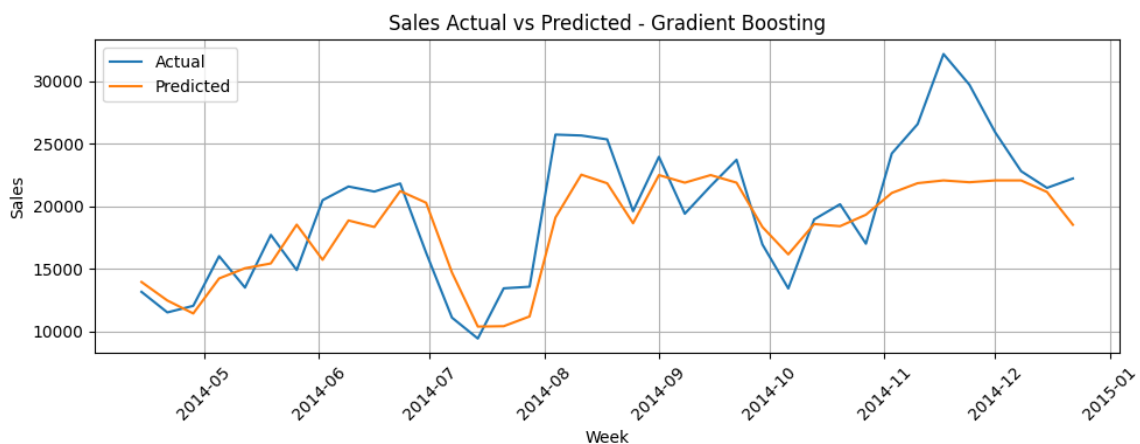
Business Features

- Price points and discount levels
- Product category attributes
- Historical performance metrics
- Promotional campaign indicators

3.3 Model Selection Rationale

Sales:

	MAE	RMSE	R2	MAPE
XGBoost	1876.059364	2609.215161	0.587627	0.127318
Gradient Boosting	1890.948210	2644.144384	0.576513	0.130679
Random Forest	1895.034298	2650.860288	0.574359	0.126579
ETS	2105.764252	2689.473107	0.561868	0.148172
Decision Tree	2171.211392	3192.308931	0.382723	0.163658
Naive	3096.132674	3608.044668	0.211478	0.209276
ARIMA(1,1,1)	6239.788279	7417.304832	-2.332441	0.372415



Quantity

	MAE	RMSE	R2	MAPE
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